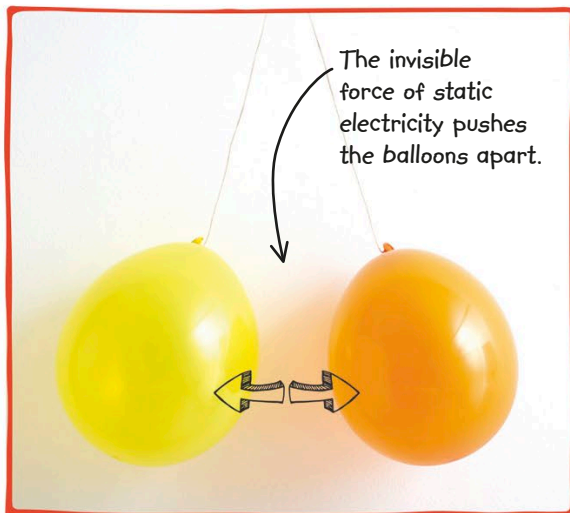


PUSHING BALLOONS

Hang two uncharged balloons together and nothing much happens. Yet, when the balloons are charged with static electricity, things start to get a little more interesting.



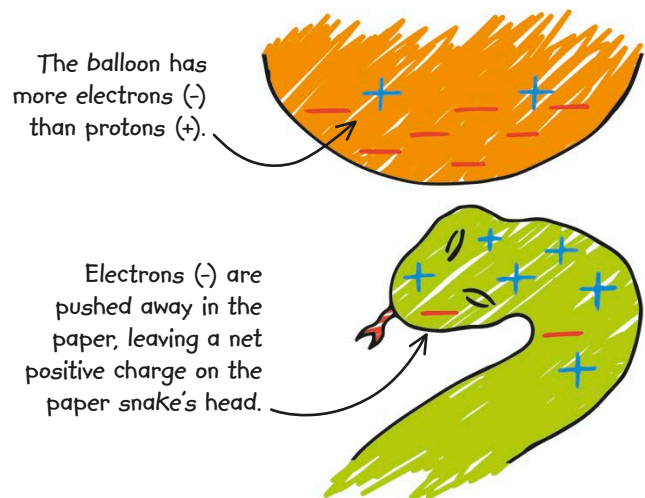
1 Tie string to each of two balloons. Do not charge them yet, but hold both strings between a finger and thumb and let the balloons hang down.



2 Charge both balloons evenly, all over their surfaces. When you dangle them down now, they stay apart, repelled (pushed) by an invisible force.

HOW IT WORKS

Electric charge is carried by tiny particles called protons, which carry a positive (+) charge, and electrons, which carry a negative (-) charge. Charges exert forces on each other: charges of the same type push apart, or repel, while opposite charges pull together, or attract. Normally, there are equal numbers of positive and negative charges everywhere. However, when you rub the balloon on wool or your hair it picks up extra electrons, giving the balloon an overall negative charge. This pushes the electrons away in the paper, making the paper's edge positively charged. That is why the paper snake is attracted to the balloon.



REAL WORLD SCIENCE LIGHTNING STORM

Inside a thundercloud, swirling winds make ice crystals in the cloud rub together, which charges them. The base of this cloud becomes negatively charged, which is attracted to a positively charged ground. This can produce lightning, which takes the shortest route to the ground, often striking trees.

